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School of Science, Computing, and Engineering Technologies

ASSIGNMENT AND PROJECT COVER SHEET

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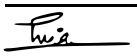
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1. Introduction

In New South Wales (NSW), the Transport for NSW (2024) report recorded 10,596 serious injuries in the 12 months ending June 2024. Understanding which conditions and locations are associated with casualty outcomes can inform where road-safety investment is most needed. The intended users of this project are local government transport planners, road safety analysts, and local council engineers. These groups can use the predictive models to identify high-risk LGAs and allocate infrastructure funding where it will have the greatest impact. The project uses historical NSW crash records (2020 to 2024) and LGA-level population estimations from the Australian Bureau of Statistics (2026).

2. Problem Framing

Road safety is an important issue because road accidents can cause injuries, deaths and property damage. Analysing crash patterns in New South Wales (NSW), Australia, can help us understand factors associated with crash severity, such as weather, road conditions, speed limits, urbanisation, seasons and LGA population. It is also important to understand how crash patterns differ between Local Government Areas (LGAs).

This project addresses three research questions: (1) Can environmental and road-condition features predict whether an individual crash will be a casualty crash? (2) Do NSW LGAs group into distinct crash-pattern profiles, and which features distinguish them? (3) Can an LGA's historical crash profile predict whether it will be a high-risk LGA in the following year?

Previous research has also explored ways to analyse and predict road fatalities. Hassouna and Pringle (2019) used ARIMA to analyse and predict road fatalities in Australia. While this research provides useful background on road fatality prediction, the current project also examines individual crash outcomes and similarities between LGAs. This provides a different perspective on road safety by considering both crash level characteristics and area level patterns.

Machine learning is suitable for this project because classification can identify patterns associated with crash casualty outcomes, while clustering can reveal groups of LGAs with similar crash characteristics. The findings may provide useful insights for future road safety analysis and planning. However, the results depend on the available data and do not prove that any particular factor directly causes a crash or makes it more likely to result in a casualty.

3. Data Collection

3.1 NSW Crash Data

The primary dataset was the nsw_road_crash_data_2020_2024_crash records published by Transport for NSW (n.d.). The original release contained 40 columns mapping individual incidents identified by a unique Crash ID. This source file was prepared in Excel before loading: the twelve crash columns were extracted in a working sheet (Sheet2) based on relevance to the research objectives. A small number of 2019 records were present in the raw file and were removed so that the crash data matched the 2020–2024 window.

3.2 ABS population data

Demographic baselines were sourced from the Australian Bureau of Statistics (2026) Regional Population release. The downloaded data cube was reduced to Table 1 (Estimated resident population, Local Government Areas) with the 2020–2024 columns retained and saved as LGA_population.xlsx. The cleaned file held 546 rows covering all Australian LGAs; 129 NSW LGAs remained after filtering.

4. Data Processing

4.1 Processing Steps and Decisions

The data pipeline transformed the records from a baseline of 92,189 raw crash rows to a final classification set of 90,857 rows. This was caused by removing 5 records from Lord Howe Island, 107 records from 2019, and 1,220 rows carrying "Unknown" category values (see Appendix A). LGA names were harmonized to match the ABS naming convention before merging, ensuring unmatched LGAs were not silently dropped during the join. The datasets were combined using a left join onto the reshaped population table, preserving every valid crash record with no missing population values. The dataset was expanded to thirteen columns in the final classification file by adding Season and Severe_Crash, merging the LGA population data, and dropping three redundant outcome-related columns to prevent target leakage.

To prepare the data for supervised learning, Severe_Crash was engineered as a binary target variable, coded as 1 for any crash involving casualties and 0 for non-casualty tow-aways. The dataset originally categorised crashes into fatal, serious, moderate, minor/other, and non-casualty tow-away outcomes. Because any injury represents a significant road safety failure, the four injury categories were aggregated to form the positive "casualty" class. For the clustering task, the data were aggregated by LGA and year to yield a balanced panel of 645 rows (129 LGAs across 5 years). The per-capita crash rate was calculated after this aggregation step, so the denominator reflected the annual population of the LGA.

4.2 Exploratory Findings

- **Class balance:** The final classification dataset consists of 67.93% casualty crashes (61,716) and 32.07% non-casualty crashes (29,141). A baseline model predicting "casualty" for all instances would yield an inflated accuracy of 67.93%, so models are evaluated using balanced accuracy, per-class F1, and PR-AUC (see Figure B1).
- **Feature association:** Univariate strength tests ranked by Cramer's V revealed weak associations: speed limit (0.0819), weather (0.0599), road classification (0.0470), and season (0.0125). All values sit below the 0.10 threshold for practical relevance (see Table B1).
- **LGA crash-rate spread:** Average annual crash rates ranged from 126.2 per 100,000 in Camden to 1,340.9 in Unincorporated NSW. A negative Spearman rank correlation ($\rho = -0.5797$, $p < 0.001$) indicated that smaller populations are associated with higher per-capita rates. t (see Figure B2).

5. Machine Learning Model Selection

5.1 Classification

For this assignment, 2 classification models were made: logistic regression.ipynb and crash casualty.ipynb to answer two research questions, those being:

1. Can environmental and road-condition features predict whether an individual crash will be a casualty crash?
2. Can an LGA's past crash profile based on crash rate, weather and road condition predict whether it will be a high-risk LGA in the following year?

Classification suits both questions because each has a binary outcome. Logistic regression was chosen as the main model because it is simple, produces probabilities and allows feature effects to be interpreted through its coefficients. For crash-level prediction, a Random Forest was also tested to compare its ability to capture non-linear relationships and feature interactions. The LGA-level model uses logistic regression only.

5.2 Clustering

The idea for our clustering model was based on two questions:

- Which NSW LGAs have similar road crash patterns?
- Which LGA year records are unusual?

These two questions were answered using two different clustering techniques, which are K-means and DBSCAN. K-means is the model that helps cluster NSW LGAs that have similar road crash patterns, where it divides records into different k groups around cluster centres to determine each groups average features of each group. However, this only assumes roughly round clusters, and it is very sensitive to outliers. With this in mind, we added DBSCAN as a second method, which identifies records that do not belong to a sufficiently dense region and labels them as noise to determine which LGS tear records are unusual. Both models were evaluated using silhouette scores (scikit-learn developers, n.d. a) (scikit-learn developers, n.d. b). To implement this, we used the processed CSV from our chosen datasets.

6. Technical Implementation

6.1 Crash severity model (crash_casualty.ipynb)

Data and target: The model uses nsw_crashes_classification.csv, containing 90,857 crashes from 2020–2024 after records with unknown values were removed. The target is 1 for a casualty crash and 0 for a non-casualty crash. Casualty crashes account for 67.93% of the dataset, providing a baseline accuracy of 67.93% for a model that always predicts a casualty crash.

Features and preprocessing: The features include weather, road classification, road surface, surface condition, urbanisation, speed limit, season and LGA population. Identifiers and variables that could reveal the outcome were excluded. Speed limit was treated as categorical, while population was standardised and categorical features were one-hot encoded within a scikit-learn pipeline.

Training and model selection: The data was split into 80% training and 20% testing, using stratification and random_state = 20.Logistic regression and Random Forest were trained with balanced class weights. Grid search with three-fold cross-validation and ROC-AUC was used for tuning. The selected settings were $C = 1$ for logistic regression and 150 trees, maximum depth 10 and minimum leaf size 20 for Random Forest.

Threshold and outputs: Decision thresholds were selected using out-of-fold training predictions to maximise balanced accuracy. The resulting thresholds were 0.51 for logistic regression and 0.50 for Random Forest. Test predictions, permutation importance and the selected model pipeline were saved for visualisation and backend use.

6.2 LGA risk model (logistic regression.ipynb)

Data and target:

The model uses 645 LGA-year records (129 LGAs × 5 years) from lga_yearly_aggregates_clustering.csv , the same file used for clustering. An LGA-year is labelled high risk if its crash rate per 100,000 residents is at or above 250 per 100,000 residents which is based off the median of the training years (254.9), and low risk otherwise. The rate is used instead of the crash count so that large LGAs are not labelled risky just because more people live there.

Features:

- Population
- severe crash
- wet-weather crash
- rural-road crash.

The crash rate, total crashes and the severe and non-severe counts were excluded since the label is baked on the median calculated from them (crash rate = total crashes ÷ population × 100,000, which the notebook confirms), so including them would let the model simply recompute the answer.

Split:

The split is by time: the model is trained from 2020 to 2023 (516 records, 50.0% high risk) and tested on 2024 (129 records, 46.5% high risk). The risk threshold comes from the training years only. The features and the label of each record come from the same LGA-year, so the model classifies LGAs in a year it has not seen rather than forecasting a future year.

Model and validation:

A scikit-learn pipeline with StandardScaler and logistic regression (random_state = 20). It was evaluated on the test set, compared with a majority-class dummy classifier, and validated with 5-fold GroupKFold cross-validation on the LGA name, so rows from the same LGA never appear in both a training fold and a validation fold. Predictions, metrics, cross-validation scores, coefficients and permutation importance are saved as CSV files for the visualisations.

6.3 Clustering model (clustering-data.ipynb)

The implementation was written as a Python notebook file using imports like pandas and scikit-learn. Here are some of the steps taken to implement this technique:

Feature Selection and Loading: The processed CSV was created based on the questions that needed to be solved, whereby the features used in this technique are:

- Crash rate per 100,000 residents
- Severe crash %
- Wet-surface crash %
- Rural-road crash %

All unincorporated NSW was not removed so that it could also be examined during the DBSCAN unusual record model creation.

Standardisation: We applied StandardScaler to all the selected features. This is necessary because the crash rates are numerical values, but the other features are percentages. Without scaling, the crash rate column would interfere with the distance calculation because of the large numerical value.

Model processing: K-means and DBSCAN were fitted with the features to solve the question stated. The resulting K-means cluster assignments and DBSCAN labels were added to a dataset. Panda import was used to calculate the average of each feature, determine pattern types of LGAs, and list all records labelled as noise. These records were saved into separate CSV files for visualisation.

7. Implementation Evaluation

7.1.1 Crash severity model (crash_casualty.ipynb)

The test set contained 18,172 crashes. Since non-casualty crashes are the minority class, precision, recall and F1 are reported for that class.

Model	Threshold	Accuracy	Precision (NS)	Recall (NS)	F1 (NS)	Balanced accuracy	ROC-AUC	PR-AUC (NS)
Baseline: always severe	n/a	0.679	0.000	0.000	0.000	0.500	0.500	0.321
Baseline: always non-severe	n/a	0.321	0.321	1.000	0.486	0.500	0.500	0.321
Logistic regression	0.51	0.568	0.372	0.502	0.427	0.551	0.569	0.378
Random Forest	0.50	0.579	0.381	0.498	0.432	0.558	0.574	0.381

Cross-validation on the training data gave a ROC-AUC of 0.569 ± 0.007 for logistic regression and 0.574 ± 0.004 for the Random Forest, and a PR-AUC (non-severe) of 0.378 ± 0.007 and 0.380 ± 0.006.

Confusion matrices (rows are the actual class, columns the predicted class):

	LR: predicted severe	LR: predicted non-severe	RF: predicted severe	RF: predicted non-severe
Actual severe	7,400	4,944	7,629	4,715
Actual non-severe	2,905	2,923	2,927	2,901

Key findings

- Weak predictive performance: Both models achieved accuracy below the 67.93% majority-class baseline. However, because balanced class weights encourage detection of the minority class, accuracy alone does not fully reflect their performance.
- Limited ability to distinguish outcomes: ROC-AUC values of approximately 0.57 and balanced accuracy values of 0.55–0.56 indicate only modest improvement over chance. The non-casualty F1 scores were also below the 0.486 achieved by the always-non-casualty baseline.
- Similar model performance: Logistic regression and performed marginally better across the reported metrics, but the differences were small. There is insufficient evidence of a meaningful performance advantage over logistic regression.

Overall, the results of this model are quite lackluster and shows that environmental and road-condition features alone are insufficient to reliably predict whether an individual crash will involve a casualty. Although both models performed slightly better than chance by metric, they are too weak to support dependable crash-level predictions. As such, we will likely use logistic regression.ipynb as the main classification model in Assignment 3 because it is simpler to interpret and much better performance. The crash casualty model is instead kept in the report as an experimental analysis of relationships between crash conditions and casualty outcomes.

7.1.2 LGA risk model (logistic regression.ipynb)

The table shows the test results on the 129 LGAs in 2024 (high risk is the positive class).

Measure	Result
Accuracy	0.74 (dummy baseline 0.535)
Precision (high risk)	0.70
Recall (high risk)	0.78
F1-score	0.74
Test ROC-AUC	0.845
Cross-validated ROC-AUC (5-fold, grouped by LGA)	0.821 ± 0.011

Confusion matrix (rows are the actual class, columns the predicted class):

	Predicted low risk	Predicted high risk
Actual low risk	49	20
Actual high risk	13	47

Key findings

- The model clearly beats the baseline. Accuracy of 0.74 compares with 0.535 for always predicting the majority class, and the cross-validated ROC-AUC of 0.82 on LGAs held out of training shows that it generalises to LGAs it has not seen. It found 47 of the 60 high-risk LGAs (13 missed) and raised 20 false alarms.
- Rural-road % is the main driver. Its standardised coefficient is +1.22, meaning a one-standard-deviation increase multiplies the odds of high risk by about 3.4, and shuffling it lowers the test ROC-AUC by 0.27.
- Population has a negative coefficient (−0.40). Larger LGAs tend to have lower crash rates. This is partly a mechanical effect, because the rate divides by population, so small LGAs have higher and less stable rates.
- Wet-weather % has a small positive coefficient (+0.33), but its effect on test performance (0.005 ~ 0.009) cannot be told apart from zero. Severe crash % has almost no effect (−0.04).

7.1.3 Comparison of the two models

The LGA model separates its classes much better (ROC-AUC 0.85) than the crash model (0.57). Each LGA record summarises hundreds of crashes, which smooths out random variation, and rural-road share is closely linked to crash rates per resident. The severity of an individual crash depends on factors that are not in the data, such as the type of crash, who was involved, the vehicles and the time of day.

7.1.4 Limitations

- Small LGAs are unstable. LGAs with a few thousand residents can switch between high and low risk because of a handful of crashes.
- Median threshold. LGAs just above or below the median are almost identical but receive different labels, so errors cluster near the boundary.
- Small test set. With only 129 test LGAs, accuracy is uncertain by roughly ±8 percentage points.

7.2.1 K-Means: Similar Crash Patterns LGA

Based on Appendix C (Silhouette score table), we choose k=3 because it has the highest silhouette score. However, all the scores are close together, which shows the clusters are not clearly separated. This is why we chose the k value with the highest silhouette score, which splits the data into three groups:

- Cluster 0: Higher rural-road %, highest casualty share, and highest casualty %
- Cluster 1: Mostly non-rural areas and low crash rate
- Cluster 2: Higher Rural-road %, wet-surface share % and highest crash rate%

kmean data	Crash rate per 100k	Severe %	Wet-weather %	Rural-road %
0	316.00	76.28	11.35	63.73
1	217.96	67.15	18.42	12.63
2	433.16	67.95	26.93	67.91

Based on the results above, it shows that rural roads are the clearest separator between Cluster 1 and the other two, because both of them are rural, while Cluster 1 is mostly non-rural. Clusters 0 and 2 are more rural; however, they differ based on the wet surface and casualty share %. Crash rate increased from Cluster 1 to Cluster 0 and was highest in Cluster 2.

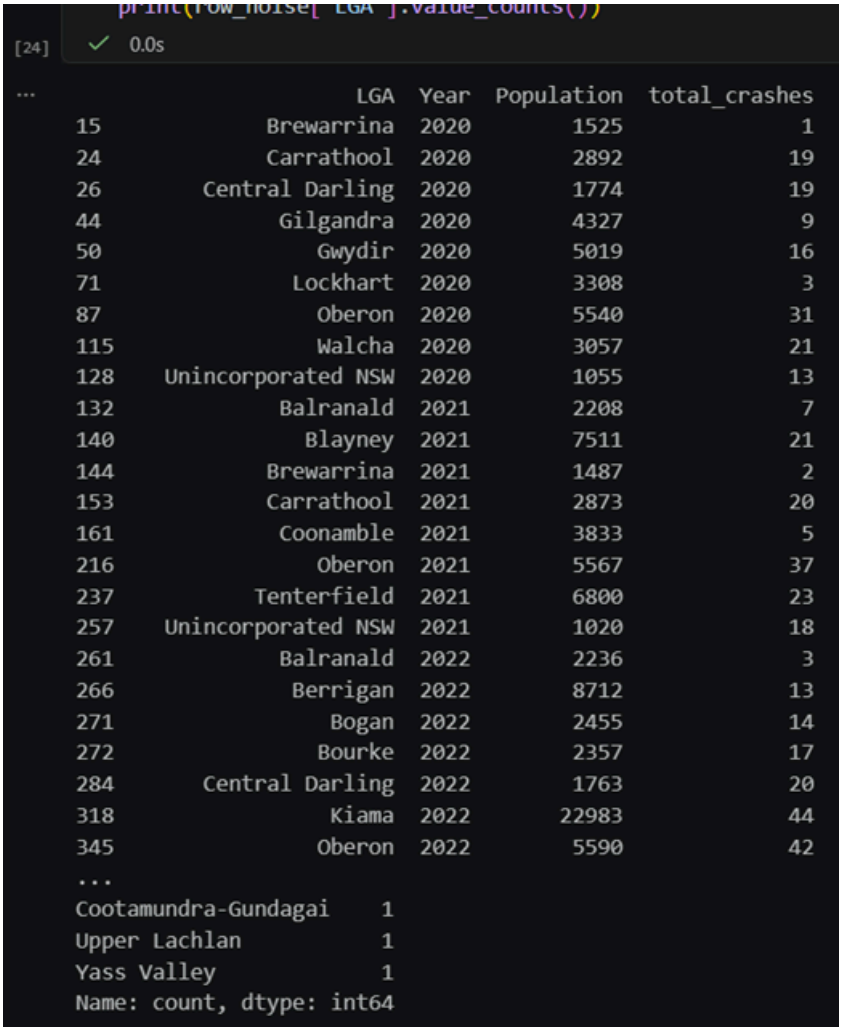
```
Cluster 0 stable LGAs: 6
Cluster 1 stable LGAs: 54
Cluster 2 stable LGAs: 4
Total LGAs: 129
```

Based on the results above, K-means was assigned each LGA year record to different clusters so it can fall in different years. This is to check how stable the clusters were. The output above shows the number of stable LGA records counted as stable per cluster. Only 64 out of the 129 LGAs stayed in one cluster in all 5 years. 54 in cluster 1, 6 in cluster 0, and 4 in cluster 2. The rest moved between clusters and had their data change throughout the years.

7.2.2 DBSCAN: LGAs Unusual records

```
eps = 0.5  Silhouette Score: -0.21  Number of noise points: 276
eps = 1.0  Silhouette Score: 0.46  Number of noise points: 41
eps = 1.5  Silhouette Score: 0.61  Number of noise points: 11
eps = 2.0  Silhouette Score: 0.64  Number of noise points: 9
```

The output above shows the noise count and silhouette score based on testing with 4 eps values (0.5,1.0,1.5, 2.0). This shows that as the eps value increases, the amount of noise and the silhouette score change. This shows that more records are merged into one cluster and leave very few noise points, which most of the time reflect a single group. We did not pick 0.5, as most of the records were labelled as noise and the silhouette score is negative. So, we chose 1.0 as it has a manageable amount of noise points while still having stable records for clusters for DBSCAN to identify a main group.



The output above shows the unusual records found by DBSCAN, which are labelled as noise. This shows that their feature values did not fit into the main group. Some of these records come from smaller LGAs with fewer crashes, as the population size is smaller. Therefore, some records may have been flagged by DBSCAN to reflect it as unstable crash rates then unusual crash patterns.

7.2.3 Limitations of K-means and DBSCAN

- Silhouette scores are kind of low, so clusters overlap slightly
- Crash rates are per resident, so they do not measure traffic volume of an LGA, which makes LGAs with small populations unstable
- Only four features were used, and the results depend on the eps value, k values, and the random state.

8. Visualisation

8.1 Charts for Clustering based on kmeans_k_scores and lga_clusters

Chart 1: Cluster Profiles

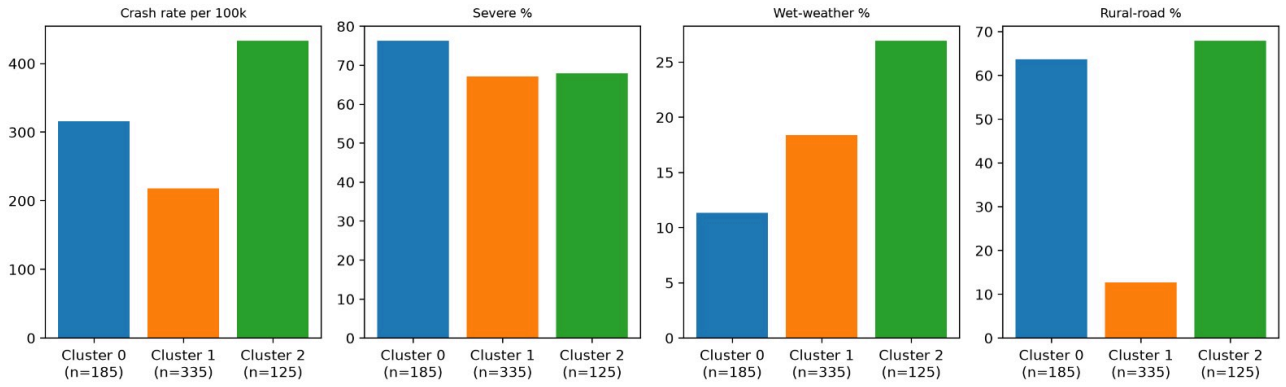


Chart 2: Scatter (PCA)

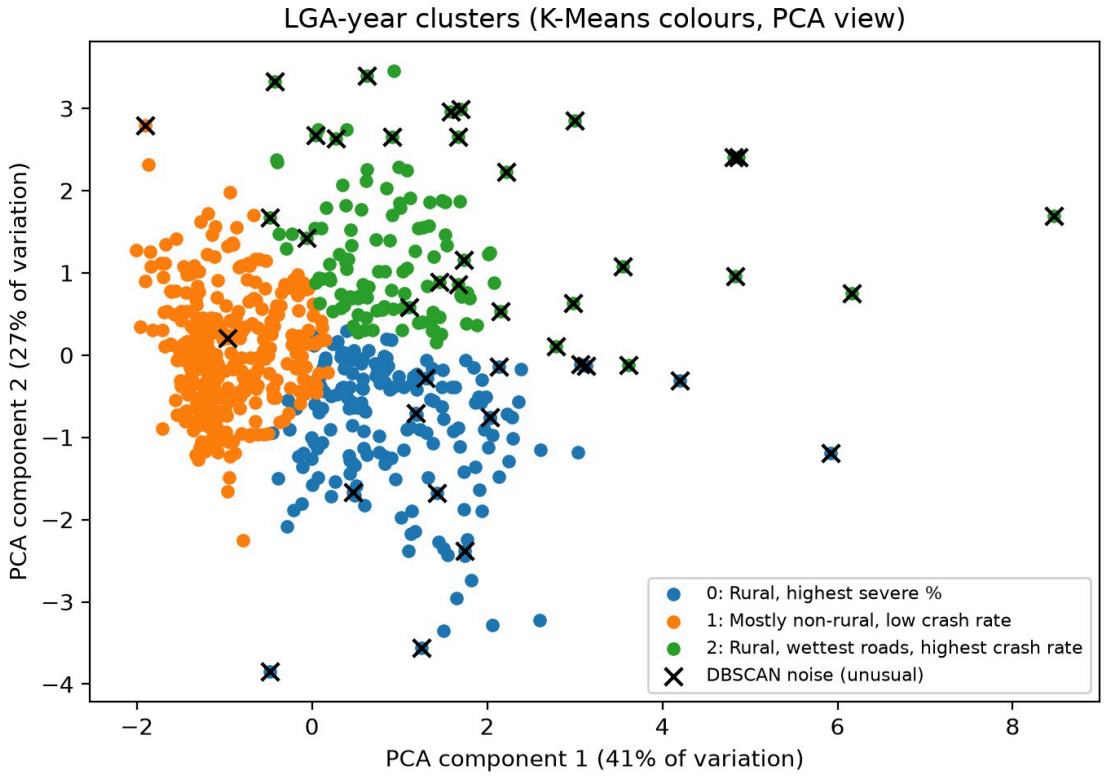
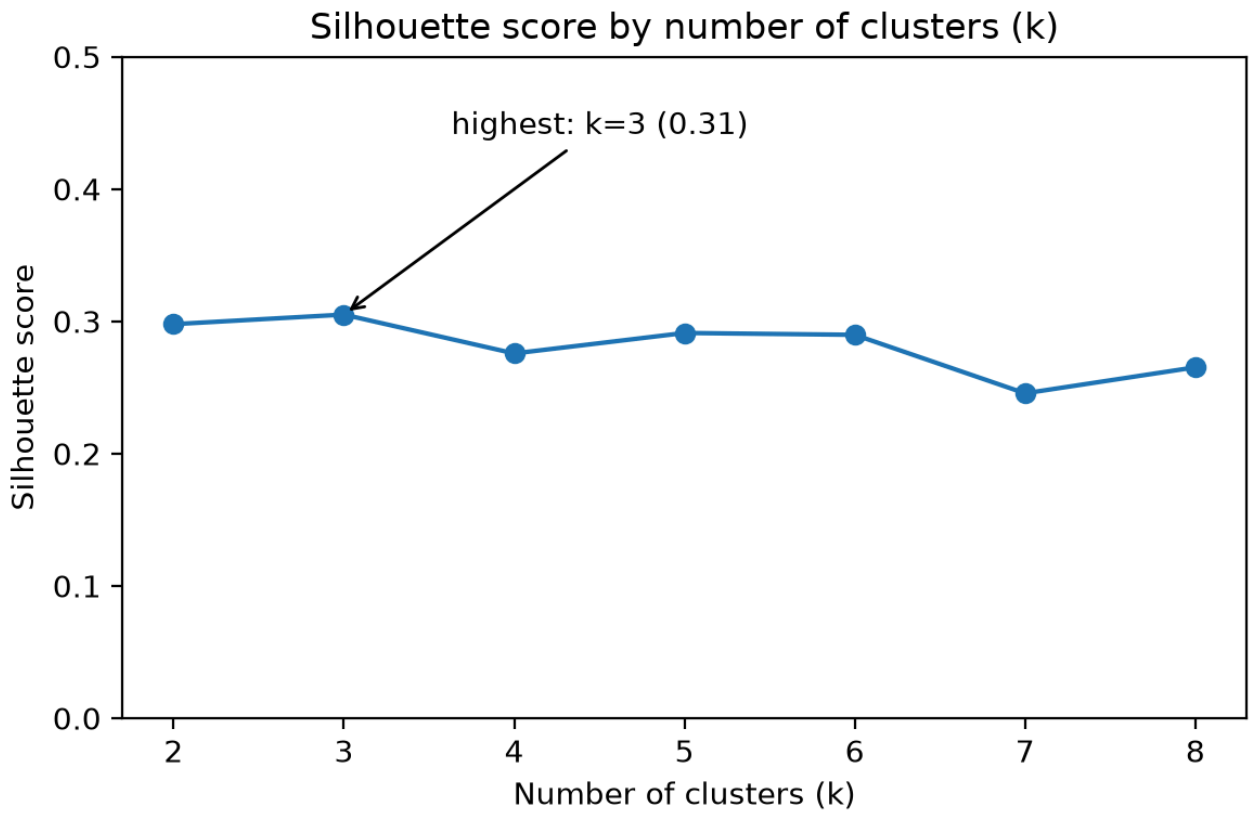


Chart 3: Silhouette by k



8.2 Charts for Classification based on logreg_permutation_importance and logreg_test_predictions

Chart 1: Class Metrics

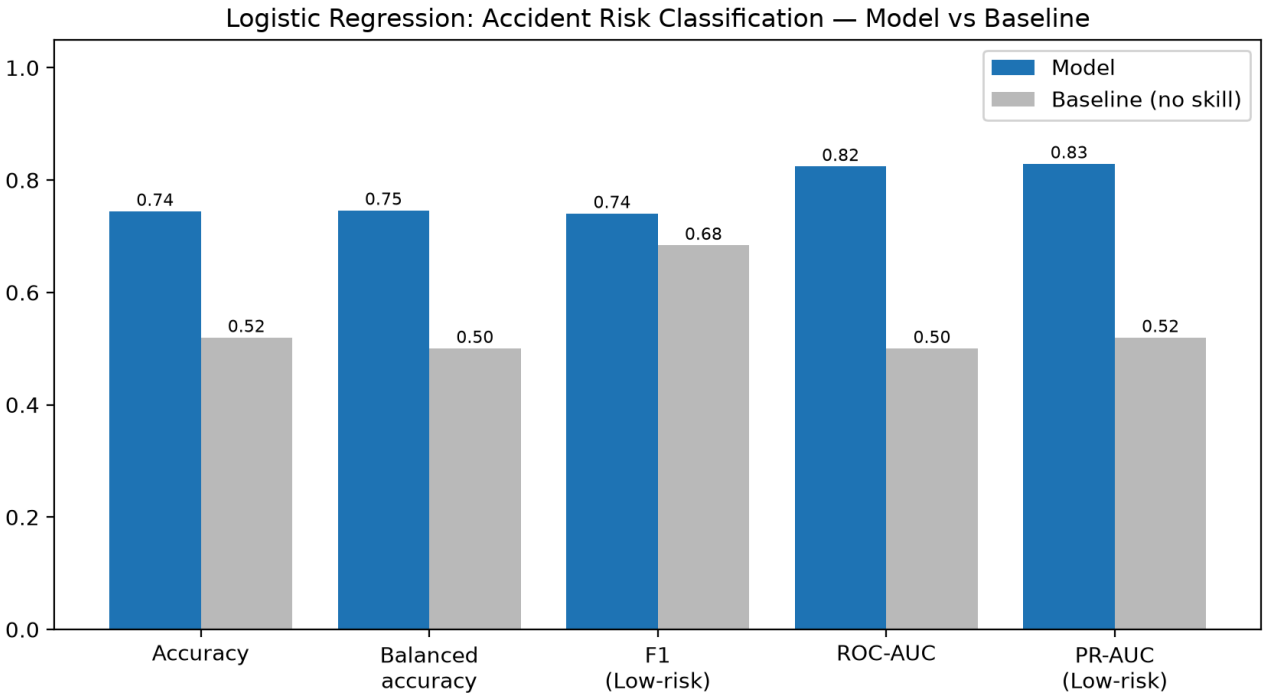


Chart 2: Confusion Matrix

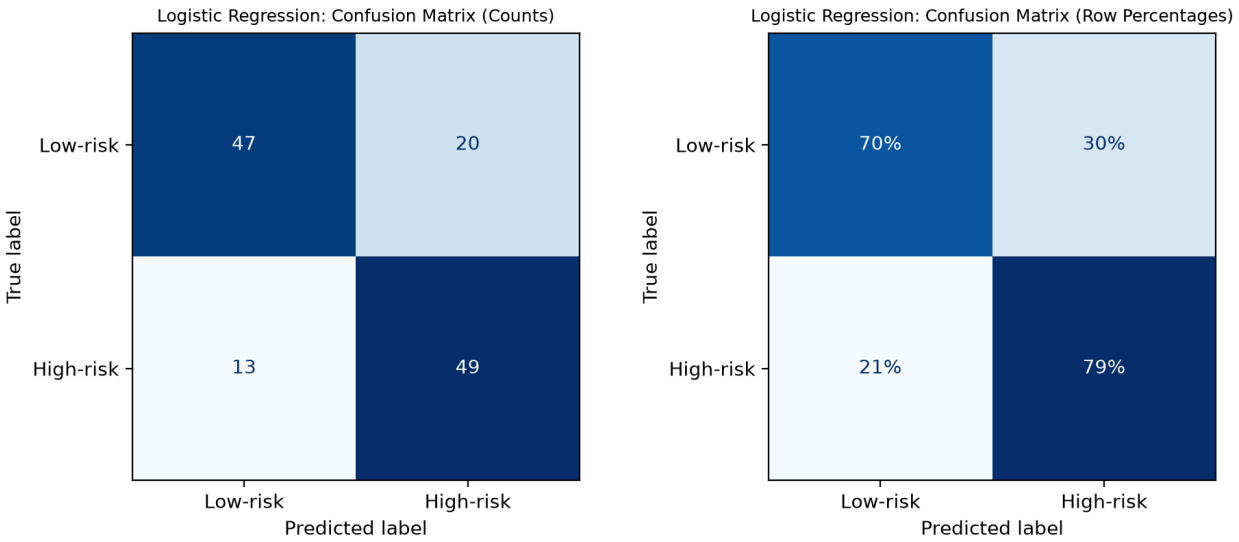


Chart 3: Feature Importance

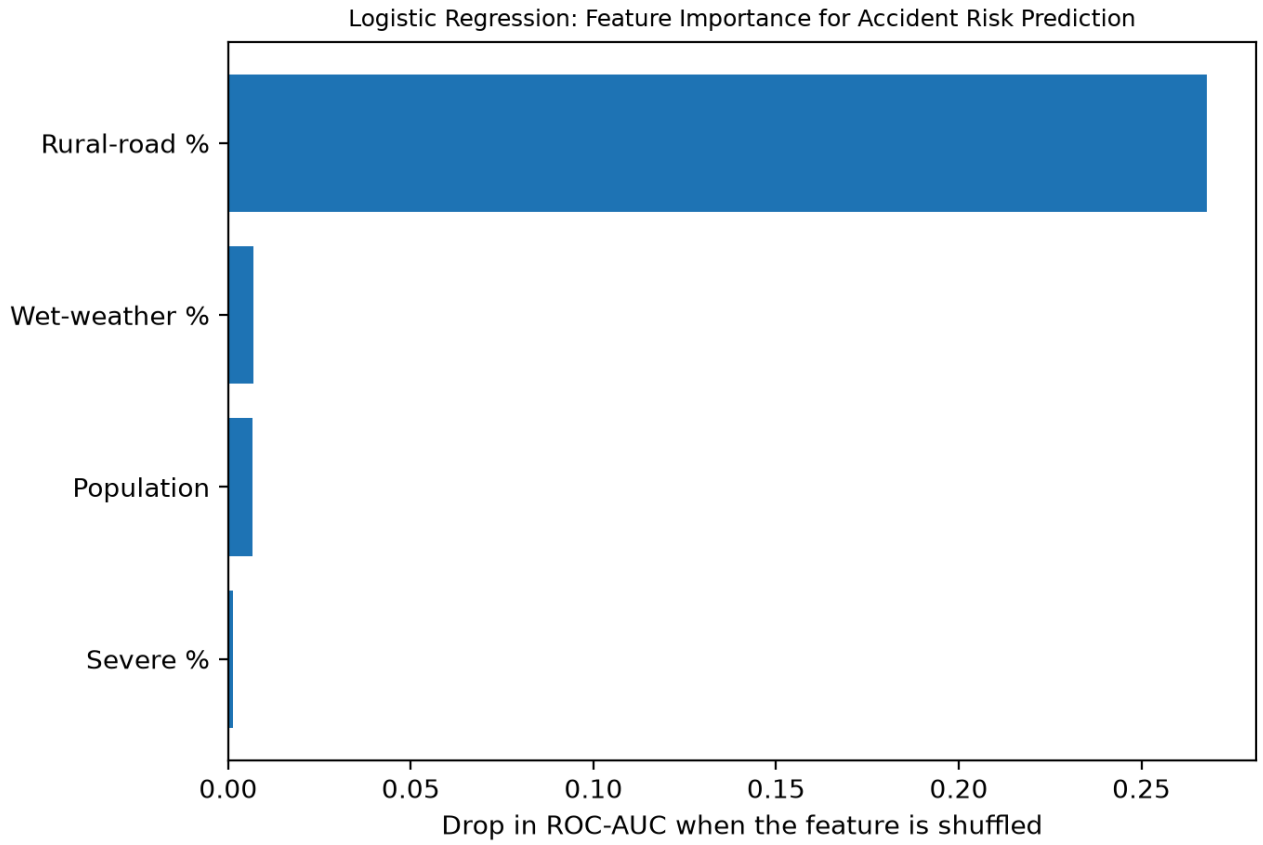


Chart 4: Probability Distribution

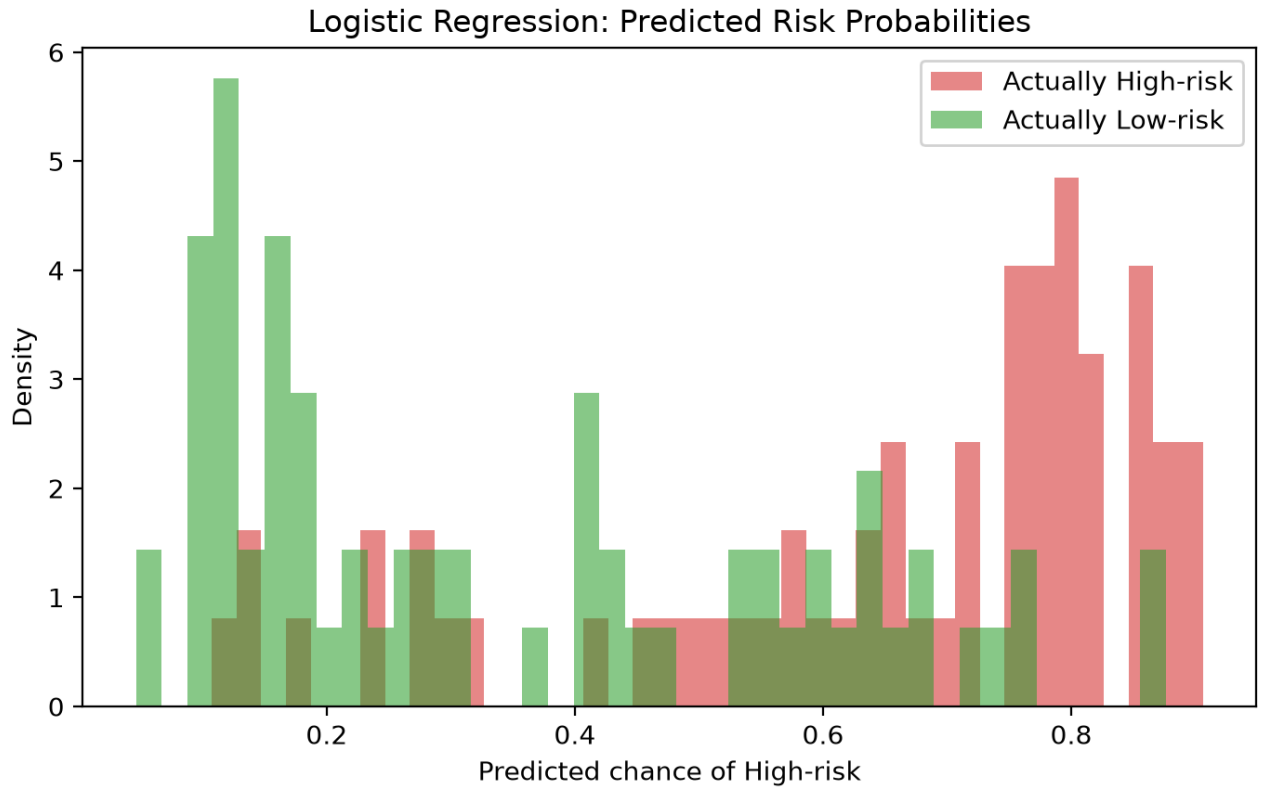
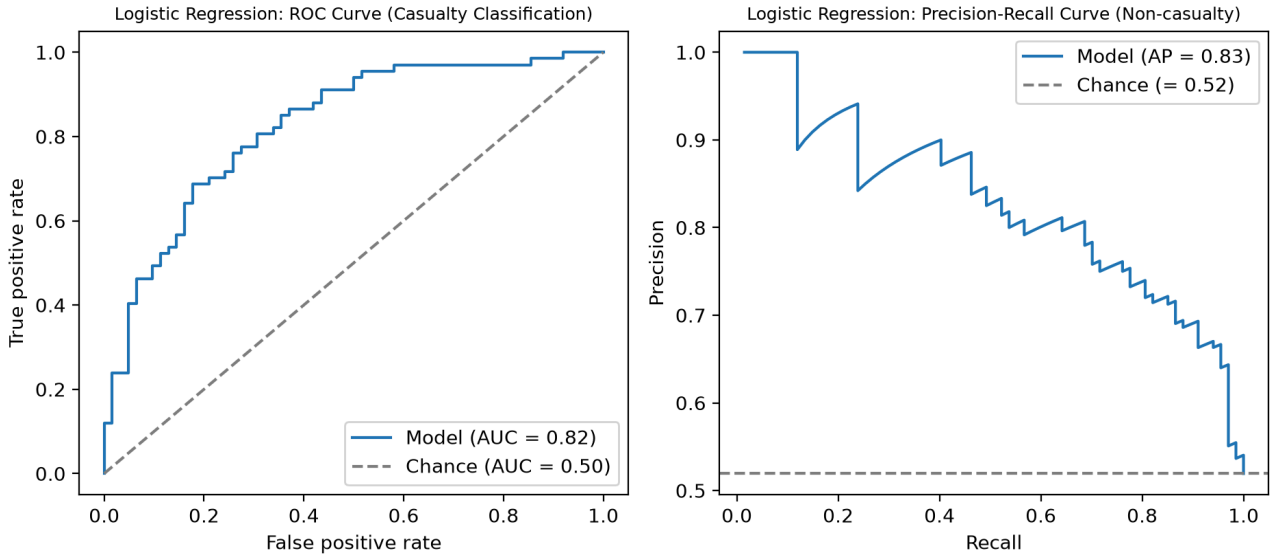


Chart 5: ROC Curve and Precision-Recall Curve



9. Conclusion

This assignment allowed us to explore how to use machine learning to analyse NSW road crash data from the year 2020 to 2024 through classification and clustering models. For classification, predicting crash casualties using the available features was challenging. However, the LGA-level model gave better results, which suggests that regional characteristics can help distinguish between different risks. For clustering, K-means identified three groups with different crash patterns, and DBSCAN showed which records did not fit well within dense groups, which showed that different regions may experience a variety of crash patterns. Overall, the project shows that machine learning can be very useful for exploring data and identifying the patterns in between them. However, the findings are limited by the data and machine learning methods used.

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Appendix A: data-processing.ipynb summary

	Step (Notebook Cell)	Rows	Columns	Effect
1	Raw crash dataset ‘Sheet2’ loaded (Cell 2)	92,189	12	Baseline crash data
2	Raw Population dataset loaded (Cell 3)	546	7	Baseline population data
3	LGA names harmonised; Lord Howe Island dropped (cell 5)	92,184	12	12 mismatches found; 11 renamed; Lord Howe Island removed (5 rows)
4	2019 crashes removed (cell 6)	92,077	12	107 rows removed; crash window set to the 2020–2024 study period.
5	NSW LGAs selected from population file (cell 7)	129	7	Non-NSW LGAs dropped
6	Population dataset reshaped from wide to long (cell 8)	645	4	One row per LGA-year
7	Crash data merged onto population (cell 8)	92,077	16	Merged the two datasets into one
8	Season added; duplicate columns dropped (cell 9)	92,077	15	Mapped seasons from month of crash
9	Severe_Crash column created (cell 10)	92,077	16	62,570 casualty / 29,507 non-casualty
10	"Unknown" rows removed (cell 11)	90,857	16	1,220 rows removed
11	Classification file written (cell 12)	90,857	13	Three redundant columns (Degree of crash, Degree of crash – detailed, month of crash) dropped
12	Clustering file written (cell 13)	645	10	129 LGAs * 5 years

Appendix B

Figure B1: Class Balance Chart

Distribution of crash severity in the final classification dataset. Severe crashes (injury or fatality) make up 67.9% of the sample; non-severe crashes (non-casualty tow-aways) make up 32.1%.

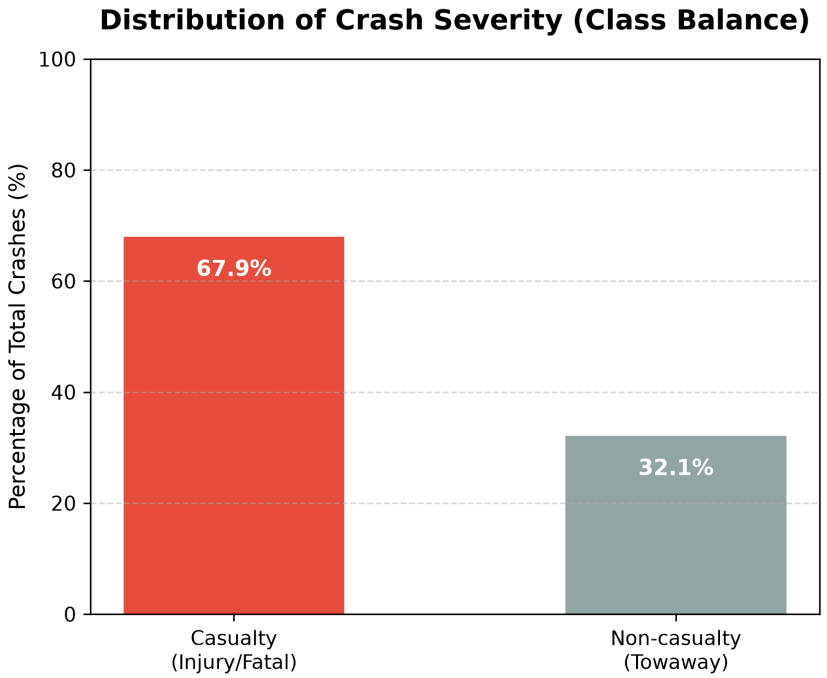


Figure B2: LGA Crash Rate Top 10/ Bottom 10

Top ten and bottom ten LGAs by average annual crash rate per 100,000 residents (2020–2024). High rates at the top are driven by small-population volatility; low rates at the bottom reflect large metropolitan populations.

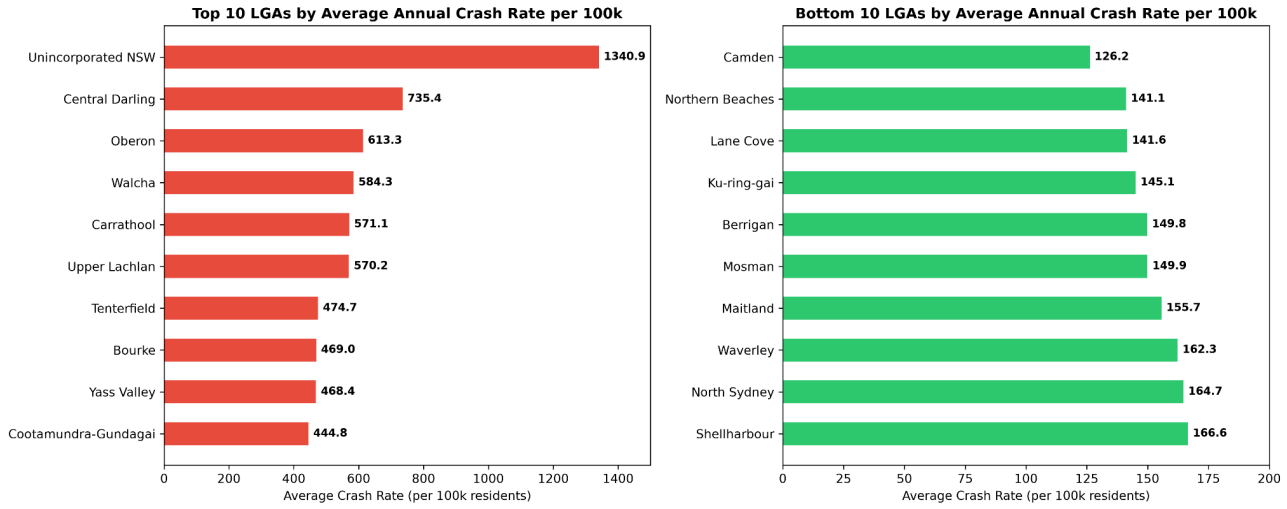


Table B1: Chi-Square and Cramér's V Association Tests

Feature	Chi-Square p-value	Cramér's V value	Strength of Association
Speed Limit	1.6337e-124	0.0819	Very Weak
Weather	2.7261e-68	0.0599	Very Weak
Road Classification	2.7131e-44	0.0470	Very Weak
Season	2.6952e-03	0.0125	Negligible

Appendix C

Silhouette score table

This table shows the silhouette scores for each k value (s in the notebook) to decide how many clusters to use.We tested K-Means with values from 2 to 8 to determine the best silhouette score with a random state of 20 which is just a random value and fixes the random seed used for initialisation

K value	Silhouette Score
2	0.30
3	0.31
4	0.28
5	0.29
6	0.29
7	0.25
8	0.27